

Add a New System Call to say hello

Target

1. Add a new system call into the linux kernel on Kylin Desktop OS v10
2. Test the new system call in user model

Tools

Install GCC Software Colletion

```
sudo apt-get install build-essential
```

How to use GCC

- [gcc and make](#)

How to do

Step0

- new customized kernel config:6.16.12
 - i. linux-6.16.12.tar.xz
 - ii. <https://cdn.kernel.org/pub/linux/kernel/v6.x/linux-6.16.12.tar.xz>

Step1 (Linux kernel 6.2+)

在文件 include/linux/syscalls.h (第1176行)

```
#endif /* CONFIG_ARCH_HAS_SYSCALL_WRAPPER */之前，添加一行：
```

```
asmlinkage long sys_schello(void);
```

Step2 (Linux kernel 6.2+)

kernel/sys.c

在文件SYSCALL_DEFINE0(gettid)函数之后（第958行），添加如下行：

```
SYSCALL_DEFINE0(schello)
{
    printk("Hello new system call schello!Your ID\n");
    print('Hello new system call schello! hello 学号\n');
    return 0;
}
```

Step3 (Linux kernel 6.2+)

针对64位OS

arch/x86/entry/syscalls/syscall_64.tbl

在文件334 common memfd_secret sys_memfd_secret 行之后，添加如下行：

335 common schello sys_schello

Step4

```
make clean
make -j5
sudo make modules_install
sudo make install
```

Step 5

重新启动：

reboot

确认新内核是否成功运行：

```
uname -a
// 注：显示和编译版本一致才为正确
```

```
hxusr@hxhost:~$ uname -a
Linux hxhost 6.5.72110731 #2 SMP PREEMPT_DYNAMIC Thu Oct 26 03:14:20 CST 2023 x86_64 x86_64 x86_64 GNU/Linux
```

Step 6

编写用户态测试程序testschello.c

```
#include <unistd.h>
#include <sys/syscall.h>
#include <sys/types.h>
#include <stdio.h>
#define __NR_schello 336
int main(int argc, char *argv[])
{
    syscall(__NR_schello);
    printf("ok! run the cmd in terminal: sudo dmesg | grep schello\n");
    return 0;
}
```

Step 7

编译用户态测试程序testschello.c，并执行

```
# 编译
gcc -o testsc testschello.c
# 运行
./testsc
sudo dmesg | grep schello
```

```
hxusr@hxhost: ~/lab05
hxusr@hxhost:~/lab05$ uname -a
Linux hxhost 6.5.72110731 #2 SMP PREEMPT_DYNAMIC Thu Oct 26 03:14:20 CST 2023 x86_64 x86_64 x86_64 GNU/Linux
hxusr@hxhost:~/lab05$ gcc -o testsc testschello.c
hxusr@hxhost:~/lab05$ ./testsc
ok! run dmesg | grep hello in terminal!
hxusr@hxhost:~/lab05$ sudo dmesg | grep schello
[ 664.762944] Hello new system call schello!Your ID
[ 708.761780] Hello new system call schello!Your ID
hxusr@hxhost:~/lab05$
```

Task (lab05)

- 实现上述实例，并输出linux kernel版本号和你的“学号”
- 撰写实验报告

End.